

Lab 02 - First Verilog

In this lab, you've learned how to do an initial and simple design in Verilog to learn the Vivado tooling and process involved in RTL/FPGA design.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

Today we learned Verilog and the process of setting up a project. We went into depth about changing settings to match the course material, manipulating files, and compiling our code to be able to be run on a board.

Lab Questions

1 - Describe the stages of building a Verilog project in Vivado.

First we create a new project with the supplied materials from our lab instructions. We loaded the simulation and made any necessary changes to the source code to get the desired output. We then simulate the design in a local environment. We compile using Generate Bitstream and push the compiled code to the board, then troubleshoot if necessary.

2 - What is the value in looking at the elaborated design schematic?

The elaborated design schematic allows us to see the components of the board we are targeting as well as the logic we are working with.

3 - Why should we simulate our designs frequently? What does the simulation do?

Simulating designs frequently allows us to troubleshoot and see where our code went wrong before the code gets to the board. The simulation shows us a virtual representation of the process that will be implemented to the board.

Code Submission

https://github.com/adamzens/ece230L_Week2.git